

Novin

Mechanical Engineer — CFD & FEA and Project Management

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Profile

Mechanical engineer with hands-on CFD experience in ANSYS Fluent (turbulence modeling, particle tracking, thermal/airflow optimization) and a strong CAD/FEA foundation. Comfortable with Linux/HPC and Python-based simulation workflows, and with the project-side deliverables that surround them: Gantt planning, ITPs, and final documentation packages. C1 English.

Work authorization: eligible to work in the EU (Luxembourg post-graduation job-search permit from autumn 2026)

Experience

Paul Wurth S.A. (SMS group)

Luxembourg, Luxembourg

Mechanical Engineer (Intern)

Feb 2026 – Jul 2026

- Developed ANSYS Fluent CFD models ($k-\omega$ SST turbulence, Lagrangian particle tracking) of an air-knife system, predicting turbulent airflow and dust transport to keep a camera enclosure clear of dust on a conveyor belt in a dusty, ambient-temperature steel-plant environment
- Ran iterative CFD design studies with mesh-independence verification to optimize air-knife geometry, improving dust-removal uniformity over the baseline design
- Designed the protective aluminium enclosure in CREO with full GD&T, balancing thermal protection, manufacturability, and serviceability
- Built and validated 3D-printed prototype iterations (Bambu Lab P1S) for fit, assembly, and functional testing

University of Luxembourg

Luxembourg, Luxembourg

Research Assistant

Apr 2025 – Nov 2025

- Simulated nanoscale heat transfer with LAMMPS molecular dynamics on a Linux HPC cluster, running and analyzing multiple simulation cases
- Automated post-processing and visualization in Python (NumPy, Matplotlib); supported Fortran-based simulation workflows

Nebraspumps

Tehran, Iran

Project Manager

Apr 2024 – Jul 2024

- Coordinated technical documentation (P&ID, WBS, ITP, test reports, commissioning plans) for industrial pump projects

Atlassarma

Tehran, Iran

Heating & Air Conditioning Engineer

Nov 2023 – Mar 2024

- Designed industrial HVAC systems (chillers, air-handling units) and sized refrigeration-cycle components
- Selected compressors and sized evaporators, condensers, and liquid lines using USA Coil, Aspen EDR, and CoolSelector
- Produced production-ready heat-exchanger models and drawings in SolidWorks

Ramin Thermal Power Plant

Ahvaz, Iran

Mechanical Engineering (Intern)

May 2018 – Aug 2018

- Assisted overhaul of shell-and-tube heat exchangers; inspected tubes, baffles, and tube sheets for scaling, fouling, and mechanical defects

Education

University of Luxembourg

Master of Engineering — Sustainable Product Creation

Luxembourg, Luxembourg

2024 – 2026

(expected Sep 2026)

- **Project Management** (coursework): project planning and scheduling (Gantt, WBS), cost control, risk management, and stakeholder communication
- **Blast Furnace Technology** (coursework): ironmaking fundamentals, blast furnace operation, instrumentation, and process monitoring
- **Master's thesis** (with Paul Wurth S.A.): CFD-driven design and optimization of an air-knife system protecting camera/sensor enclosures of dust on a conveyor belt (ambient-temperature steel-plant setting) — ANSYS Fluent, $k-\omega$ SST turbulence, Lagrangian particle tracking
- **Design project**: full machine-design calculations and topology optimization of a single power-screw press in Fusion 360, FEA-verified (static structural / stress analysis), reducing part mass by 30% while meeting strength requirements
- Additional simulation in ANSYS (FEA): steady-state thermal and modal analysis; 3D modeling, assemblies, and connection calculations in Autodesk Inventor; additive manufacturing (Prusa FDM)

Jundi Shapur University of Technology (Dezful)

Bachelor of Engineering — Mechanical Engineering

Dezful, Iran

2016 – 2020

- **Final project**: thermal and hydraulic design of a shell-and-tube heat exchanger to cool hot turbine oil using river water as the coolant in a power-plant application — sizing and analysis in Aspen and ANSYS
- CFD & FEA in ANSYS (Fluent, static structural) for fluid flow, heat transfer, and structural analysis
- CAD/CAE: SolidWorks, CATIA, AutoCAD (3D modeling, assemblies, technical drawings)

Skills

CFD & Simulation

- ANSYS Fluent: RANS CFD, $k-\omega$ SST turbulence modeling, Lagrangian particle tracking, meshing & mesh-independence, post-processing
- ANSYS Mechanical (FEA): static structural, steady-state thermal, and modal analysis
- LAMMPS molecular dynamics; Aspen EDR & USA Coil (heat-exchanger / thermal sizing)

CAD & Mechanical Design

- SolidWorks, CREO, Autodesk Inventor (Nastran), Fusion 360 (generative design), AutoCAD
- GD&T, design-for-manufacturability, technical drawings, additive manufacturing (FDM 3D printing)

Programming & Computing

- Python (NumPy, Matplotlib), MATLAB, Bash
- Linux & HPC cluster environments; familiar with Fortran workflows

Language Proficiency

- **English**: C1-certified by IELTS
- **French**: B1 proficiency
- **Persian**: Native / Bilingual